

CYBER SECURITY COURSE

Windows Live Response and Initial Compromise

Professional Case Report 01

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Case ID	CSC-2026-01
Discipline	Digital forensics
Overall risk	High
Evidence status	Confirmed by saved Redline evidence

Purpose

Identify the first reliable indicators of compromise and establish a defensible starting point for the incident timeline.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

A Windows endpoint contained suspicious files in C:\ProgramData\kav. The directory name imitated antivirus software and required validation through MFT metadata, hashes, signatures and file attributes. The directory and file metadata provide a credible initial lead. The unsigned svchost.exe copy and the Tools.zip MFT record should be treated as attacker-controlled artifacts until disproved.

Case Profile

Scenario	A Windows endpoint contained suspicious files in C:\ProgramData\kav. The directory name imitated antivirus software and required validation through MFT metadata, hashes, signatures and file attributes.
Environment	Windows forensic image reviewed with Mandiant Redline and Lab 1.mans evidence collection.
Tools	Mandiant Redline, NTFS MFT metadata, MD5 comparison, Digital signature review

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Loaded the Redline collection and isolated C:\ProgramData\kav in File System evidence.
2. Compared Standard Information and Filename timestamps.
3. Reviewed hashes, file sizes, attributes and signature state.
4. Tagged suspicious artifacts and recorded them as indicators of compromise.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	C:\ProgramData\kav\svchost.exe	High	Created 2017-06-13 02:57:08 UTC; MD5 d9cac44b7b505237767951ec4138eaf4; unsigned binary using a trusted Windows filename outside the system directory.
F02	C:\ProgramData\kav\Tools.zip	Medium	MFT INDEX record with MD5 d41d8cd98f00b204e9800998ecf8427e; evidence indicates a deleted or empty archive record remained in NTFS metadata.
F03	Masquerading directory	High	The kav directory is not a standard Windows directory and plausibly imitates Kaspersky naming to reduce suspicion.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

Redline File System filter: Full Path contains C:\ProgramData\kav

Hash validation: MD5 and signature status

Risk Assessment

Risk	Level	Operational impact
Unauthorized executable	High	Execution, persistence and follow-on tool deployment
Masquerading	Medium	Delayed detection by users and administrators

Recommendations

1. Quarantine the endpoint and preserve the Redline collection.
2. Block the recorded hashes and search for the same paths across other endpoints.
3. Acquire the binaries for controlled malware analysis.
4. Alert on svchost.exe executions outside Windows system directories.

Knowledge Demonstrated

- Live response triage
- MFT interpretation
- Hash-based analysis
- IOC documentation

Conclusion

The directory and file metadata provide a credible initial lead. The unsigned svchost.exe copy and the Tools.zip MFT record should be treated as attacker-controlled artifacts until disproved.

Hands On Lab Practical 01

Completed as part of the authorized cyber security course practical work.